BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 43

Note: This section consists of 43 part questions and all are to be answered. Each question carries ONE mark. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 2 Hours — Maximum Marks: 85 (As per condensed syllabus)

Q1. Choose the correct answer for each from the given options:

Qi. The volume of 3.01×10^{23} molecules of H_2 at S.T.P will be:

(A) $11.2dm^3$ (B) $22.4dm^3$

(C) $33.6dm^3$ (D) $44.8dm^3$

Qii. 602.10 has:

(A) 3 significant figures (B) 4 significant figures

(C) 5 significant figures (D) 6 significant figures

Qiii. One mole of H_2O contains number of hydrogen atoms:

(A) 6.02×10^{23} (B) 1.204×10^{24}

(C) 3.01×10^{23} (D) 1.204×10^{23}

Qiv. The phenomenon is due to surface tension:

(A) Density (B) Vapour Pressure

(C) Viscosity (D) Capillary Action

Qv. The rate of diffusion of CO_2 is equal to:

(A) CO (B) CH_4

(C) C_2H_4 (D) SO_2

Qvi. This law is applied in collection of gases over water:

(A) Graham's Law of Diffusion
(B) Dalton's Law of Partial Pressure
(C) Avogadro's law
(D) Charle's law

Qvii. Volume-volume relationship is based on:

(A) Avogadro's Law
(B) Gay-Lussac's Law
(C) Dalton's Law
(D) Boyle's Law

Qviii. In James Chadwick's experiment α-particles were bombarded on:

(A) Beryllium
(B) Boron
(C) Bromine
(D) Carbon

Qix. This colour has the largest wavelength in the visible spectrum:

(A) Red
(B) Violet
(C) Blue
(D) Green

Qx. The maximum number of electrons in a particular sub-energy level is:

(A) n^2
(B) $2n^2$
(C) $2l + 1$
(D) $2(2l + 1)$

Qxi. In they hydrogen atom spectrum, the series of line obtained when electron jumps from higher orbits to the first orbit, is called:

(A) Lyman Series
(B) Balmer Series
(C) Paschen Series
(D) Brackett Series

Qxii. By emitting α-particle, $_{92}U^{238}$ converts into:

(A) $_{82}Pb^{208}$
(B) $_{84}Po^{210}$
(C) $_{89}Ac^{227}$
(D) $_{90}Th^{234}$

Qxiii. The range of bond energy of hydrogen bond is:

(A) 10 – 20 kJ/mole
(B) 20 – 40 kJ/mole
(C) 40 – 50 kJ/mole
(D) 50 – 60 kJ/mole

Qxiv. 1 Debye is equal to:

(A) 3.335×10^{-30} Cm
(B) 6.02×10^{-23} Cm
(C) 9.11×10^{-27} Cm
(D) 1.602×10^{-19} Cm

Qxv. This molecule has the highest bond energy but the shortest bond length:

(A) C = C
(B) C ≡ C
(C) O = O
(D) N ≡ N

Qxvi. This molecule has zero dipole moment:

(A) NO_2 (B) CO_2

(C) SO_2 (D) H_2O

Qxvii. The octet rule is not applied on this molecule:

(A) N_2 (B) F_2

(C) O_2 (D) H_2

Qxviii. This molecule has linear structure:

(A) CH_4 (B) NH_3

(C) BF_3 (D) C_2H_2

Qxix. The number of bonds in ethyne (C_2H_2) is:

(A) Two Sigma, Two Pi (B) Three Sigma, Three Pi

(C) Three Sigma, Two Pi (D) Two Sigma, Three Pi

Qxx. This is not an intensive property:

(A) Volume (B) Surface Tension

(C) Viscosity (D) Boiling Point

Qxxi. At constant volume heat absorbed by a system is:

(A) $q_r = \Delta E$ (B) $q_p = \Delta H$

(C) $q = E + PV$ (D) $W = P\Delta V$

Qxxii. In this reaction Kp = Kc is:

(A) $H_2 + I_2 \rightleftharpoons 2HI$ (B) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

(C) $2NO_2 \rightleftharpoons N_2O_4$ (D) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

Qxxiii. In an exothermic reaction, increase of temperature favours:

(A) Forward direction (B) Reverse direction

(C) Remain at Equilibrium (D) Irreversible

Qxxiv. The reaction $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ is called:

(A) Neutralization (B) Hydrolysis

(C) Esterification (D) Electrolysis

Qxxv. According to principle of Le-Chatlier the yield of the products cannot be controlled with the help of:

(A) concentration (B) pressure

(C) catalyst (D) temperature

Qxxvi. When this salt is dissolved in water it does not hydrolyse:

(A) NaCl
(B) CH_3COONa
(C) Na_2CO_3
(D) $NaHCO_3$

Qxxvii. This cation has the greatest tendency to get hydrated:

(A) Na
(B) Mg
(C) Al^{+3}
(D) K

Qxxviii. This aqueous solution is basic:

(A) NH_4Cl
(B) NaCl
(C) Na_2CO_3
(D) KCl

Qxxix. The oxidation number of N in HNO_3 is:

(A) +2
(B) +3
(C) +5
(D) +7

Qxxx. This is considered as weak electrolyte:

(A) HCl
(B) HNO_3
(C) NaOH
(D) CH_3COOH

Qxxxi. The order of photo chemical reaction is:

(A) zero
(B) first
(C) second
(D) third

Qxxxii. A powdered zinc is more reactive than a chunk of zinc due to:

(A) Higher Tempeature
(B) Greater Volume
(C) Greater Surface Area
(D) Higher Pressure

Qxxxiii. This compound does not contain hydrogen bond:

(A) CH_4
(B) H_2O
(C) NH_3
(D) HF

Qxxxiv. Motor Oil grading is based on:

(A) Surface Tension
(B) Viscosity
(C) Vapour Pressure
(D) Boiling Point

Qxxxv. The colour of universal indicator in neutral solution is:

(A) Red
(B) Blue
(C) Pink
(D) Green

Qxxxvi. Two solid compounds having the same crystal structure are called:

(A) Allotropes
(B) Isomorphous
(C) Polymorphous
(D) Isotopes

Qxxxvii. The particle having mass 1836 times that of electron is:

(A) Proton
(B) Neutron
(C) Meson
(D) Hyperon

Qxxxviii. This particle having mass 1836 times that of electron is:

(A) 760 torr
(B) 700 torr
(C) 720 torr
(D) 760 atm

Qxxxix. This one is not same in Na^{+} (Z = 11), Mg^{+2} (Z = 12) and Al^{+3} (Z = 13):

(A) number of shells
(B) number of electrons
(C) electronic configuration
(D) number of protons

Qxl. The (n + l) value for 4f orbital is:

(A) 4
(B) 5
(C) 6
(D) 7

Qxli. This pair of compound has the same empirical formula:

(A) C_2H_4 and C_2H_2
(B) CH_3 and C_3H_6
(C) HCHO and CH_3COOH
(D) C_3H_6 and C_3H_6

Qxlii. This molecule have two lone pair and two bond pair electrons:

(A) NH_3
(B) CCl_4
(C) H_2O
(D) $BeCl_2$

Qxliii. The change of concentration per unit time is called the:

(A) rate of reaction
(B) order of reaction
(C) rate constant
(D) equilibrium constant

Section 'B' (Short-Answer Questions)

Maximum Marks: 42

Note: Attempt FIVE questions from this section. All questions carry equal marks.

Q2.

(i) Define the following: ☆ Significant figure ☆ Viscosity ☆ Catalyst ☆ Molar Volume

(ii) Write the postulates of Kinetic Molecular Theory of gases.

(iii) An organic compound producing air pollution contains 8.73% Carbon, 77.45% Chlorine and 13.82% Flourine. Find the molecular formula of the compound if its molecular mass is 137.5 amu.

(iv) If 53.5g of NH_4Cl is heated with $Ca(OH)_2$, how many grams of NH_3 is produced? Also find the volume of NH_3 at S.T.P according to the following equation. $2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + CaCl_2 + 2H_2O$

(v) Calculate ΔHf for the following data: ☆ $6C + 3H_2 \longrightarrow C_6H_6$, ΔHf = ? ☆ $H_2 + ½O_2 \longrightarrow H_2O$, ΔHf = -286kJ/mol ☆ $C + O_2 \longrightarrow CO_2$, ΔHf = -394kJ/mol ☆ $C_6H_6 + 15/2\ O_2 \longrightarrow 6CO_2 + 3H_2O$, ΔHf = -3267kJ/mol

(vi) State Pauli's Exclusion Principle. Write the electronic configuration of the following: ☆ Ca^{++} (Z = 20) ☆ Cr (Z = 24) ☆ Br (Z = 35)

(vii) Given scientific reason for any four of the following: ☆ A falling drop of liquid is spherical in shape ☆ A liquid is less viscous at high temperature ☆ Water has high boiling point than HF. ☆ Evaporation is a cooling process. ☆ Milk sour more rapidly in summer than in winter.

(viii) 9.2 of ethyl alcohol and 12g of acetic acid kept at constant temperature until equilibrium was established 4.0g of acid was left unused. Calculate Kc for the reaction: $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

(ix) State first law of thermodynamics and show that: ☆ qp = ΔH ☆ W = PΔV

(x) Discuss the effect of following factors on rate of reactions. ☆ Concentration ☆ Radiation

(x) (OR) 1.40dm^3 volume of gas collected at a temperature of 27°C and a pressure of 900torr was found to have a mass of 2.273g. Calculate molecular mass of the gas.

Section 'C' (Detailed-Answer Questions)

Maximum Marks: 17

> Note: Attempt any One question from this section. All questions carry equal marks.

Q3.

(a) What are cathode rays? Describe the experiment of their discovery. Write their properties.

(a) (OR) Write the postulates of Bohr Atomic Theory. Derive an expression for radius of nth orbit in hydrogen atom.

(b) Balance the following equations by ion electron method: ☆ $Cr_2O_7^{2-} + I_2 \longrightarrow Cr^{3+} + IO_3^{1-}$ Acidic medium ☆ $MnO_4^{1-} + SO_3^{2-} \longrightarrow Mn^{+2} + SO_4^{-2}$ Basic medium

Q4.

(a) Write the postulates of VSEPR Theory. Explain the shape of the molecules which have: ☆ three bond pair electron ☆ two bond pair and two lone pair

(b) Differentiate between: ☆ Sigma Bond and Pi Bond ☆ Line spectrum and Continuous spectrum ☆ Hydration and Hydrolysis

Q5.

(a) Write the postulates of Arrhenius Theory of Ionization calculate Oxidation Number of the following: ☆ Cr in $Cr_2O_7^{--}$ ☆ S in $H_2S_2O_7$ ☆ N in NH_4^+

(b) State the law of mass action. Derive Kc for general reversible reaction and predict the extent of reaction.

(b) (OR) What is Natural Radioactivity? Write the properties of α and β particles.

> نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔